Methods matter: exploring the 'too much, too soon' theory, part 1: causal questions in sports injury research

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ABSTRACT

Background It is widely accepted that athletes sustain sports injury if they train 'too much, too soon'. However, not all athletes are built the same; some can tolerate more training than others. It is for this reason that prescribing the same training programme to all athletes to reduce injury risk is not optimal from a coaching perspective. Rather, athletes require individualised training plans. In acknowledgement of athlete diversity, it is therefore essential to ask the right causal research question in studies examining sports injury aetiology.

Purpose In this first part of a *British Journal of Sports Medicine* educational series, we present four different causal research questions related to the 'too much, too soon' theory and critically discuss their relevance to sports injury prevention.

Content If it is true that there is no 'one size fits all' training programme, then we need to consider by how much training can vary depending on individual athlete characteristics. To provide an evidence-base for subgroup-specific recommendations, a stronger emphasis on the following questions is needed: (1) *How much training is 'too much' before athletes with different characteristics sustain sports-related injury?* and (2) *Does the risk of sports injury differ among athletes with a certain characteristic (eg, high experience) compared with athletes with other characteristics (eg, low experience) depending on how much training they perform?*

Conclusion We recommend that sports injury researchers aiming to examine the 'too much, too soon' theory should carefully consider how they, assisted by coaches, athletes and clinicians, pose their causal research question. In the light of the limitations of population-based prevention that intends to provide all athletes with the same advice, we argue that a stronger emphasis on research questions targeting subgroups of athletes is needed. In doing so, researchers may assist athletes, clinicians and coaches to understand what training advice/programme works best, for whom and under what circumstances.

INTRODUCTION

Athletes, coaches, clinicians and researchers commonly agree that athletes sustain sports-related injury because they train 'too much, too soon'. Although this theory is widely accepted in a practical/clinical setting, we need to consider the evidence base underpinning it. While doing so, the question remains if the too much, too soon theory is based on too little.

Shedding light on the 'too much, too soon' theory requires careful consideration of the type of research question being asked and investigated. In the context of sports medicine, causal research questions help scientists and clinicians to understand and explain why sports injury occurs.[1 2] Such knowledge is essential in which to design, implement and evaluate logical injury prevention initiatives.[3] To produce prevention-oriented findings relevant for clinicians, coaches and athletes, formulating appropriate research questions is a critical first step of the scientific process.[1 4] Like methods matter in sports injury research,[5] we also emphasise that research questions matter in much the same way!

In the present educational review, which is the first part in a *British Journal of Sports Medicine* series on methods in relation to the 'too much, too soon' theory, we present four different causal research questions and we critically discuss their relevance for sports injury prevention.

CRITICAL—AND OFTEN OVERLOOKED—DISTINCTION: ATHLETE-SPECIFIC OR POPULATION-BASED PREVENTION?

When formulating a causal research question, we encourage researchers and clinicians to consider the following distinction: is your goal to prevent injuries in specific athletes or in the wider population of athletes? In the latter case, generalised population-based prevention has the aim of providing everyone with the *same* advice. For instance, policymakers encourage everyone to avoid smoking to improve public health (ie, there are no exceptions to this rule). Likewise, in a sports injury context, cycling federations recommend all athletes to wear a helmet to reduce the risk of skull fractures. Unfortunately, it's not so easy in the area of training and sports injury as not all athletes should be provided with the same advice on training. Read on as we explain why!

DIFFERENT CAUSAL QUESTIONS

As sports injury occurs due to an imbalance between the load applied to a structure (eg, training exposure measured as training amount, training load or workload) and its capacity to withstand that load,[6–10] many sports injuries are considered as overuse

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Box 1 Training exposure can be measured as 'training amount', 'training load' and/or 'workload'

This *British Journal of Sports Medicine* methods series will refer to the training exposure as either training amount, training load or workload. These terms have different meanings and they have been defined differently, depending on the source.[7 10 11] Consequently, we defined them for this paper and for the methods series as a whole.

Training amount
The number of repetitions (eg, lifts, steps, throws, swings or revolutions), the distance covered and/or the duration of a training session undertaken by the athlete.

Training load
The musculoskeletal structure-specific cumulative load applied during a training session. The training load depends on the number of repetitions (load repetition mechanism), the magnitude of load applied per repetition (load magnitude mechanism) and the distribution of load over tissue structure applied per repetition (load distribution mechanism). Importantly, two training sessions of equal training amount (number of repetitions/distance/duration) may result in vastly different training loads for the musculoskeletal structures involved.

Workload
The cumulative amount of stress placed on an individual athlete from multiple training sessions and games/competition over a specified period of time.[10] As the term 'stress' is somewhat ambiguous, and as workload has been used to cover a range of vastly different training exposures in the sports injury literature,[10 11] we will limit the use of workload in the series.

injuries. In other words, injury will occur if an athlete trains 'too much, too soon'.[8 11] This is widely accepted and has been well-known across athletic settings and the sports injury health sciences for many years.[11 12] Consequently, training amount or training load is a central component to consider when posing a causal research question in relation to sports injury occurrence. The International Olympic Committee stated: 'there can be no 'one size fits all' training or competition programme. Ultimately, the time frame of recovery and adaptation—and hence susceptibility to injury—varies within and among athletes'.[11] If this is true, personalised, athlete-specific injury prevention advice is needed. Therefore, the causal research questions in this domain should be formulated based onthe premise that athletes are different, which aligns with a personalisedinjury prevention approach.

In table 1, we present four different causal questions and their link to either population-based public health prevention or personalised athlete-specific prevention. We encourage the reader to note that the first two questions are related to average/indirect/direct causal effects in the population, ultimately leading to generalised population-based prevention. The latter two questions are concerned with subpopulation differences, which allows for injury prevention in subgroups of athletes, hence a more 'personalised' (although this still deals with subgroups) approach. There are other questions that can be asked in addition to those framed in the table. However, it is beyond the scope of the present article to discuss other causal questions.

WHY POPULATION-BASED PREVENTION MAY BE UNSUITABLE

Causal questions 1 and 2 may be unsuitable in sports injury research. Imagine that a higher training amount (eg, 15 km running/week) associates with a greater risk of sports injury compared with a lower training amount (eg, 10 km running/week) even after adjustment for all possible confounders. From a public health perspective, this would be valuable as the number of sports injuries can theoretically be reduced by recommending everyone to reduce training amount to 10 km/week. Clearly, such advice could be highly inappropriate for some subpopulations as they may be able to tolerate higher training amounts than others without being at a high risk of injury. Figure 1 illustrates this scenario.

Population A: experienced athletes with previous injuries

If the causal question is to investigate the causal effect of a training amount variable on sports injury occurrence (eg, question 1 or 2, table 1) following adjustment for confounders, then the experienced athletes with previous injuries (equivalent to population A in figure 1) would be recommended to limit their training amount to 10 km/week, which may be associated with fewer health-related benefits. Nevertheless, restricting the training amount to 10 km/week in population A is sensible advice overall as it is associated with a low to moderate injury risk (ie, close to the yellow zone).

Population B: experienced athletes without previous injuries

Experienced athletes without previous injuries (population B in figure 1) are recommended to reduce training amount to 10 km/week even though they already have limited injury risk at 15 km/week. This advice is counterproductive to population B's performance and/or health-related goals as the athletes are recommended to run less than they can.

Table 1 Different causal questions and their link to sports injury prevention (for an in-depth description of the differences between average, direct, in-direct and total causal effect, we guide the reader to complementary literature[19 20])

Number	Causal question	Link to sports injury prevention	Advice
1	What is the average causal effect of a training exposure variable on sports injury occurrence?	Population-based prevention	Same advice for everyone
2	What is the direct, indirect and/or total causal effect of other variables (eg, scapular control) than training exposure on sports injury occurrence?	Population-based prevention	Same advice for everyone
3	How much training is 'too much' before athletes with different characteristics (eg, experience) sustain sports-related injury?	Personalised athlete-specific prevention	Different advice to different athletes
4	Does the risk of sports injury differ between athletes with a certain characteristic (eg, high experience) compared with athletes with other characteristics (eg, low experience), depending on how much training they perform?	Personalised athlete-specific prevention	Different advice to different athletes

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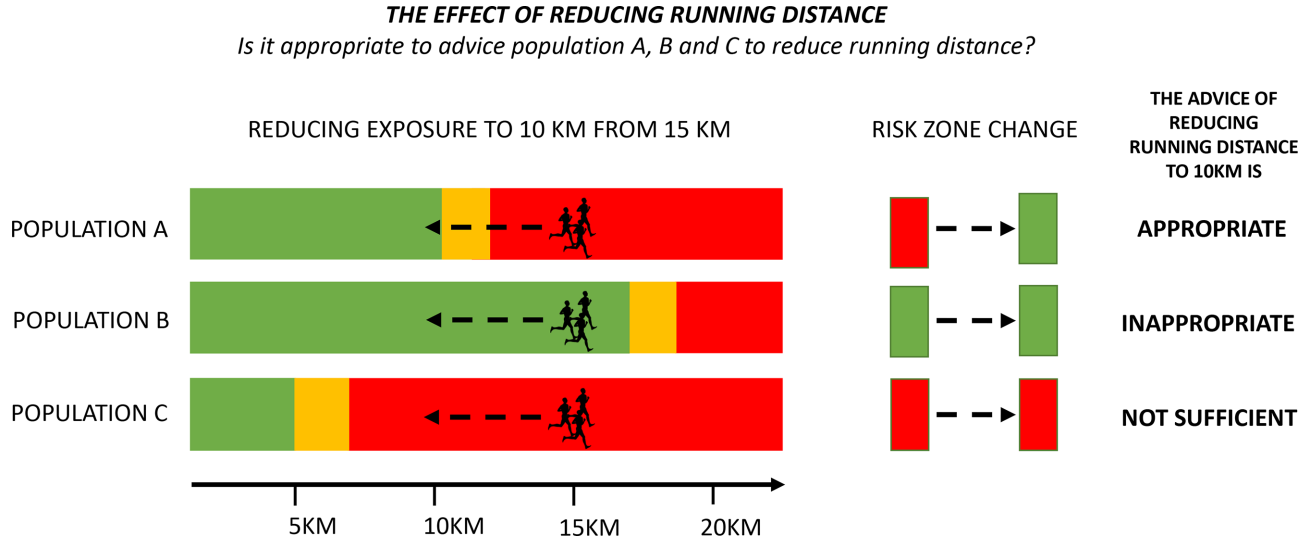

Figure 1 Examining the association between running distance (15 km/week vs 10 km/week) in three different subpopulations of athletes. Population A includes experienced athletes with previous injuries. Population B includes experienced athletes without previous injuries. Population C includes non-experienced athletes with previous injures. Green indicates low risk of injury. Yellow indicates moderate injury risk. Red denotes high risk of injury. If the population (which consists of group A,B,C) is included as a confounder, we are likely to reduce injury risk on a population level. However, we also advise population B to run less than it can, and we still advise population C to run more than it is able to before being at high risk of injury. With these considerations in mind, we question if it makes sense to estimate the average effect of change in running distance from 15 km/week to 10 km/week when including population as a confounder. Note: the content of the figure is only used for concept illustration.

Population C: non-experienced athletes with previous injures

Finally, non-experienced athletes with previous injures (population C in figure 1) are still exposed to 'too much' (ie, retained in the red zone), which may be disruptive for their continued commitment to sports participation. They should be recommended to run less than 10 km/week.

The aforementioned three populations should be prescribed with their own tailored training amount. Indeed, to avoid recommending subpopulation B to reduce its activity level (although they are able to remain at a higher level), and to avoid recommending subpopulation C to reduce to a level which is still associated with high injury risk, alternative solutions than investigating average/indirect/direct causal effects (transferring to causal questions 1 and 2 in table 1) in the total population (ie, A+B+C) are needed.

DIFFERENT TRAINING AMOUNT IN DIFFERENT ATHLETES: ATHLETE-SPECIFIC ADVICE

In causal question three, we ask, 'How much training is 'too much' before athletes with different characteristics sustain sports-related injury? The key in this question is the emphasis on training amount and the interplay with the characteristics of the athletes. The primary exposure of interest is training amount (eg, 10 km/week vs 15 km/week), whereas athlete characteristics are considered as effect-measure modifiers that literally *modify* the effect of training amount on sports injury risk. In taking this approach, the effect of training amount on sports-related injury is prioritised, and the extent of this effect is differentiated among athletes with different characteristics. The beneficiaries of this question are athletes or coaches, as they are interested in knowing how much training (and/or change in training) should be *prescribed* based on individual characteristics. Although there are many possible biological and behavioural characteristics to choose from, the researcher must be deliberately selective about which (modifiable) effect-measure modifiers are used. To see examples of the use of effect-measure modification, we guide the reader to publications elsewhere, for example, from a handball context[13] or a running context.[14] In the third piece in this methodology series on the 'too much, too soon' theory, we will provide an in-depth description of the concepts of effect-measure modification and confounding.

The end goal of question 3 is to inform personalised injury prevention (ie, sub-population advice rather than generalised advice to everyone). This is done by providing athletes with specific characteristics different training advice/programme. A training exposure-focused intervention is particularly relevant when dealing with athletes who have characteristics that cannot be manipulated and is highly attractive to coaches who want the best performances from their athletes/teams. This line of reasoning is comparable to arguments posed against the 'one-drug-fits-all' method as the effect of a drug may differ across different patients.[15] In sports injury research as in other research domains,[16] the core mission is to understand which training advice/programme works best, for whom and under what circumstances.

SAME TRAINING AMOUNT IN DIFFERENT ATHLETES

In causal question 4, we ask, 'Does the risk of sports injury differ between athletes with a certain characteristic (eg, high experience) compared with athletes with other characteristics (eg, low experience) depending on how much training they perform?' This time, the primary exposure is the athletes' characteristic (eg, experience) and the effect-measure modifier is the training exposure. Here, we are interested in the effect of an athlete characteristic on sports-related injury risk and whether this effect is different at different training exposures. The beneficiaries of this question are those designing training programmes (eg, a coach for a handball team planning the same preseason for all players or an organisation making a start-to-run programme) as they want to know which athletes are at increased risk of injury when following their specific programme (which can be different if the coach/organisation exposes the team/athletes to a different programme). This question investigates which athletes are better

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suited to follow a certain programme and/or which athletes should change modifiable factors to face a lower injury risk when following a certain programme.

DISCUSSION

Scientific practice is generally driven by research questions and hypotheses rather than by the statistical approach.[17] As an example of questionable practice, we regret to see that some sports injury researchers are guided by p values (eg, by using stepwise approaches), not by their assumptions. When designing a sports injury research study, researchers must be explicit about the goal of the research.[17] If the goal is causal inference, researchers need to clearly explain their causal assumptions[18] and proceed to ask a research question that can provide direct answers to guide injury prevention activities. which is focused on *either* a population or individual/subgroup level.[4] We emphasise that it is not wrong to examine average/direct/indirect/total causal effects in sports injury research (eg, questions 1 and 2 in table 1) if the goal is population-based prevention. In this case, the sports injury researcher is encouraged (if not required) to disclose their causal assumptions prior to any analyses.[18–20]

CONCLUSION

We recommend that sports injury researchers aiming to examine the 'too much, too soon' theory should carefully consider how they, assisted by coaches, athletes and clinicians, pose their causal research question. In the light of the limitations of population-based prevention that intends to provide all athletes with the same advice, we argue that a stronger emphasis on research questions targeting subgroups of athletes is needed (eg, numbers 3 and 4 in table 1). In doing so, researchers may assist athletes, clinicians and coaches to understand what training advice/programme works best, for whom and under what circumstances.

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